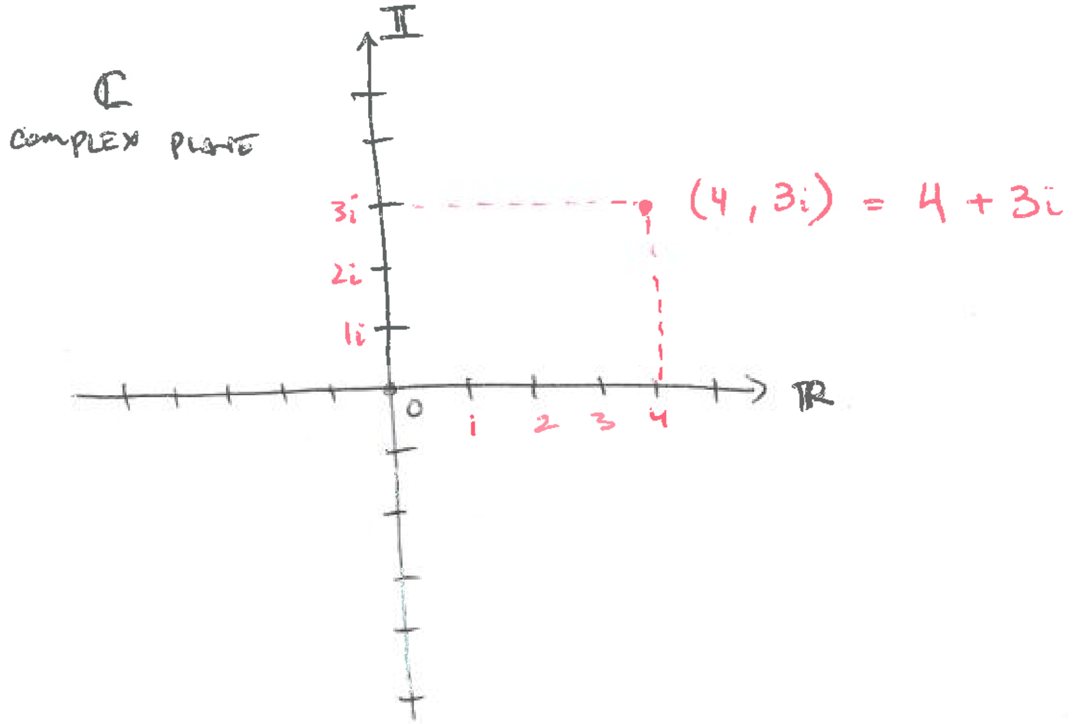
COMPLEX NUMBERS



ANY COMPLEX # $z = x + iy$

COMPLEX CONJUGATE $(*)$

$$z^* = x - iy$$

↑ CHANGES SIGN

e.g. $z = 4 + 3i$, $z^* = 4 - 3i$

$z = -4 - 3i$, $z^* = -4 + 3i$

$Re(z)$, $z = x - iy \Rightarrow Re(z) = x$

$$Re(z) = \frac{z + z^*}{2}$$

e.g. $z = 4 + 3i$

$$Re(z) = \frac{(4+3i) + (4+3i)^*}{2}$$

$$= \frac{4 + \cancel{3i} + 4 - \cancel{3i}}{2}$$

$$= \frac{8}{2} = 4$$

$\mathrm{Im}(z)$, $z = x + yi$ $\quad \mathrm{Im}(z) = y$

$$\mathrm{Im}(z) = \frac{z - z^*}{2i}$$

eg. $z = 4 + 3i$

$$\mathrm{Im}(z) = \frac{(4+3i) - (4+3i)^*}{2i}$$

$$= \frac{4+3i-(4-3i)}{2i}$$

$$= \frac{6i}{2i}$$

$$= 3$$

ADD

$z_1 + z_2 =$ $\qquad z_1 = (a+bi) \quad z_2 = (c+di)$

$$= (a+bi) + (c+di)$$

$$= (a+c) + (b+d)i$$

$$z_1 + z_2 = z_2 + z_1$$

MULTIPLICATION

$$z_1 z_2 = (a+bi)(c+di)$$

$$= ac + adi + bci + bdi^2$$

$$= ac + adi + bci - bd$$

$$= (ac - bd) + (ad + bc)i$$

$$z_1 z_2 = z_2 z_1$$

DIVISION

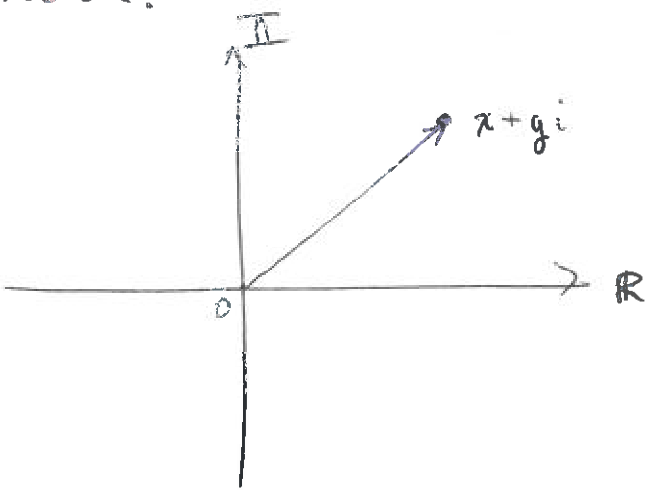
$$\frac{z_1}{z_2} = \frac{(a+bi)}{(c+di)} \qquad z = x + yi$$

$$\frac{z_1 z_2^*}{z_2 z_2^*} = \frac{a+bi}{c+di}\,\frac{(c-di)}{(c-di)}$$

$$= \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$

$$= \underbrace{\left(\frac{ac+bd}{c^2+d^2}\right)}_{\text{real}} + \underbrace{\frac{(bc-ad)i}{c^2+d^2}}_{\text{imaginary}}$$

NORM.



vector

$$|v| = (v \cdot v)^{1/2}$$

complex

WRONG

$$\cancel{|z| = (z \cdot z)^{1/2}}$$

$$\cancel{((x+yi)(x+yi))^{1/2}}$$

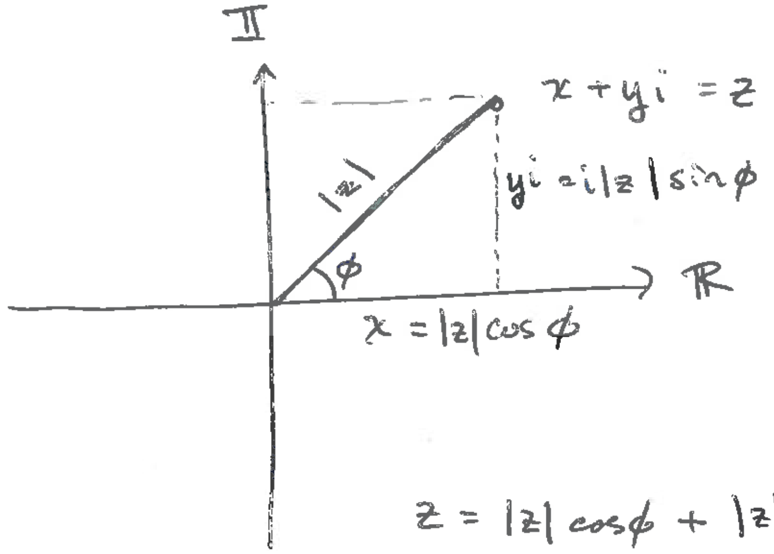
$$\cancel{(x^2 - y^2 + 2xyi)^{1/2}}$$

$$|z| = (z \cdot z)^{1/2} = (zz^*)^{1/2}$$

$$= ((x+yi)(x+yi)^*)^{1/2}$$

$$= ((x+yi)(x-yi))^{1/2}$$

MAGNITUDE $|z| = (x^2+y^2)^{1/2}$



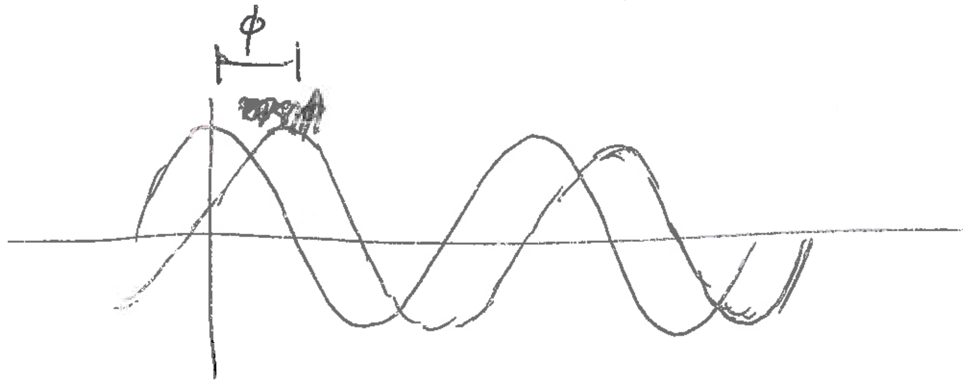
$$z = |z|\cos\phi + |z|\, i \sin\phi$$

$$= |z|(\cos\phi + i\sin\phi)$$

$|z| = (x^2 + y^2)^{1/2}$ — AMPLITUDE

$$\phi = \tan^{-1} \frac{\mathrm{Im}(z)}{\mathrm{Re}(z)}$$

$\phi = \tan^{-1} \frac{y}{x}$ — PHASE



e.g. $z^* = \left(|z|(\cos\phi + i\sin\theta)\right)^*$

$$= |z|(\cos\phi - i\sin\phi)$$

$$\cos\phi = 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \cdots$$

$$i\sin\phi = i\phi - \frac{i\phi^3}{3!} + \frac{i\phi^5}{5!} - \cdots$$

$$\cos\phi + i\sin\phi = 1 + i\phi - \frac{\phi^2}{2!} - \frac{i\phi^3}{3!} + \frac{\phi^4}{4!} + \frac{i\phi^5}{5!} - \cdots$$

$$e^{i\phi} = \cos\phi + i\sin\phi$$

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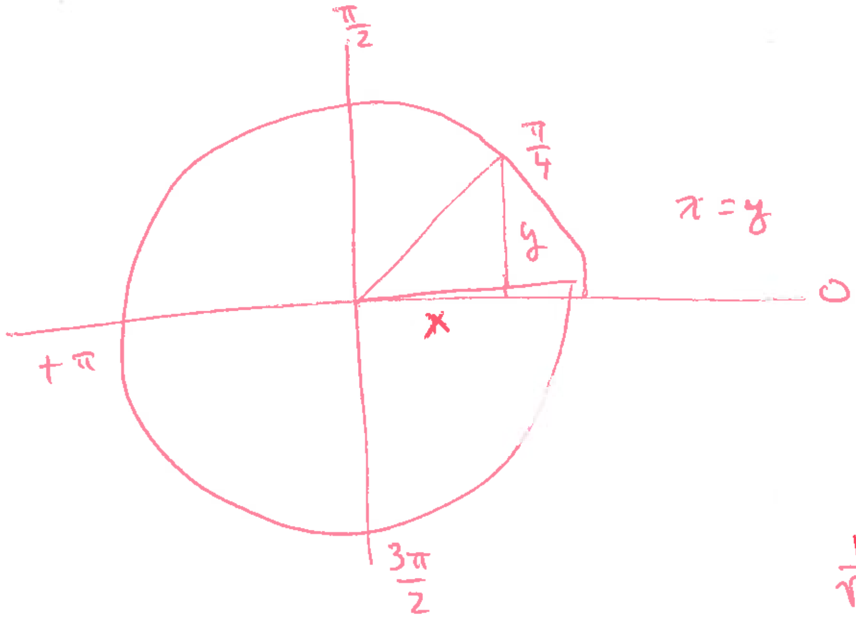
eg $e^{i0} = e^0 = 1$

$$e^{i\frac{\pi}{4}} = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$$



$$1^2 = x^2 + y^2$$
$$1^2 = x^2 + x^2$$
$$1^2 = 2x^2$$
$$\frac{1}{2} = x^2$$
$$\sqrt{\frac{1}{2}} = x$$
$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} =$$
$$\frac{\sqrt{2}}{2} =$$

$$\boxed{e^{i\frac{\pi}{4}} = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}}$$

$$e^{i\frac{\pi}{2}} = 0 + i1$$
$$= i$$

$$e^{i\pi} = -1 + 0i$$
$$\boxed{e^{i\pi} = -1}$$

EULER'S FORMULA

$$z = |z| (\cos\phi + i \sin\phi)$$

$$z = |z| e^{i\phi}$$

TAYLOR SERIES

$$f(z) = f(a) + \sum_{n=1}^{\infty} f^{(n)}(a) \frac{(z-a)^n}{n!}$$

FACTORIAL $\quad n! = 1 \cdot 2 \cdot 3 \cdot \ldots n$

$a = 0, \quad f(z) = \exp(z) = e^z \qquad z \in \mathbb{R}$

$$e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{4!} + \frac{z^5}{5!} + \ldots$$

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \ldots$$

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \ldots$$

$$e^{i\phi} = 1 + i\phi + \frac{(i\phi)^2}{2!} + \frac{(i\phi)^3}{3!} + \frac{(i\phi)^4}{4!} + \frac{(i\phi)^5}{5!} + \ldots$$

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

$$i^5 = i$$

$$e^{i\phi} = 1 + i\phi - \frac{\phi^2}{2!} - \frac{i\phi^3}{3!} + \frac{\phi^4}{4!} + \frac{i\phi^5}{5!} + \ldots$$

$e^{in\pi}$ n; even 1

n odd -1

$$E_x(z) = E_0 \exp\left(-\frac{(1+i)}{\sqrt{2}} z\right)$$ z, depth

$$= E_0 \exp\left(-\frac{1}{\sqrt{2}} z - \frac{i}{\sqrt{2}} z\right)$$

$$= \underbrace{E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right)}_{\text{MAGNITUDE}} \exp\left(\underbrace{-\frac{i}{\sqrt{2}} z}_{\text{PHASE}}\right)$$

$$= E_0 \exp\left(-\frac{1}{\sqrt{2}} z\right)\left[\cos\left(-\frac{1}{\sqrt{2}} z\right) + i \sin\left(-\frac{1}{\sqrt{2}} z\right)\right]$$

$$\boxed{E_x(z) = E_0 \underbrace{\exp\left(-\frac{1}{\sqrt{2}} z\right)}_{\text{EXPONENTIAL DECAY}} \underbrace{\left[\cos\left(\frac{1}{\sqrt{2}} z\right) - i \sin\left(\frac{1}{\sqrt{2}} z\right)\right]}_{\text{CIRCLE}}}$$